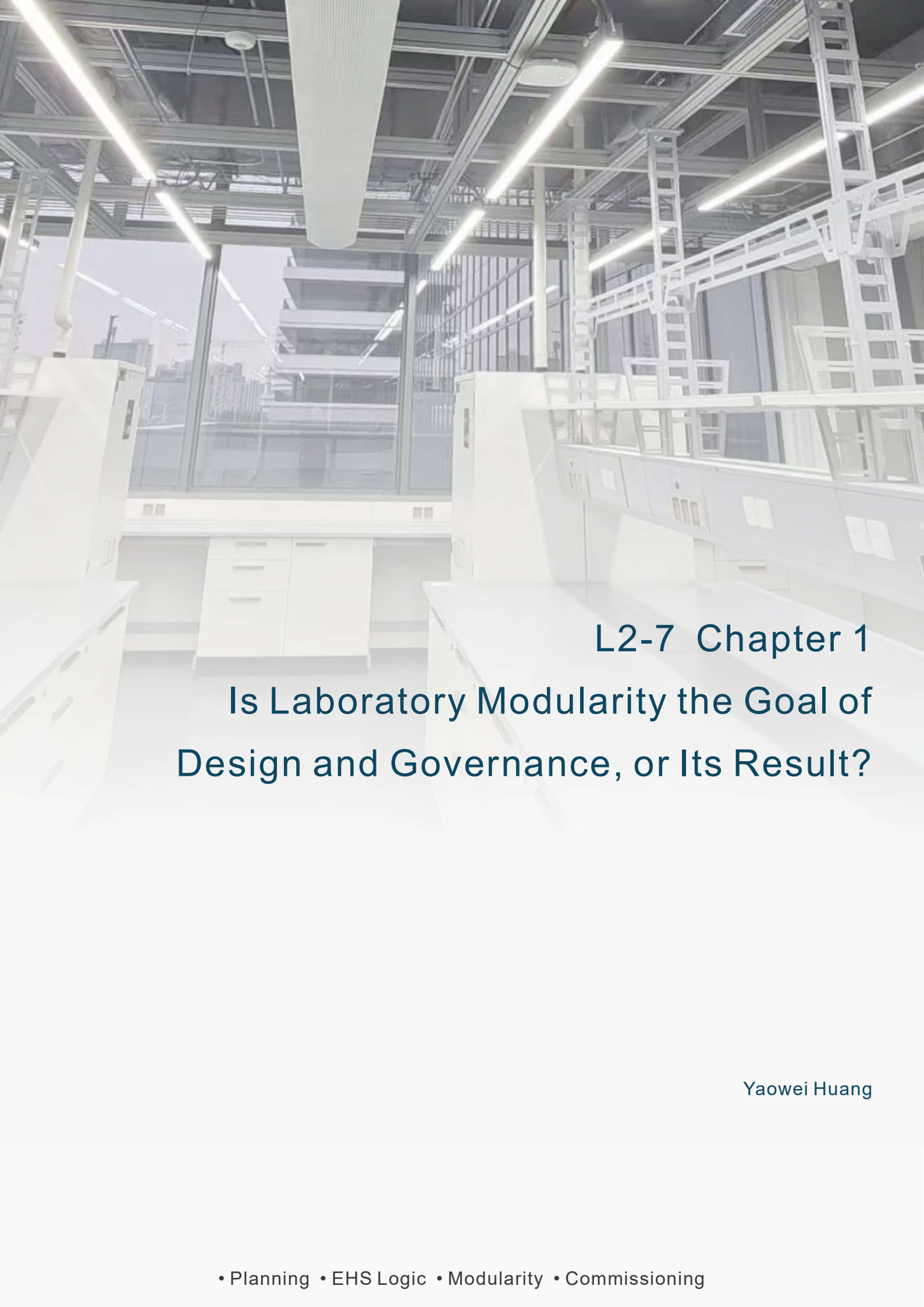




L2-7 Chapter 1

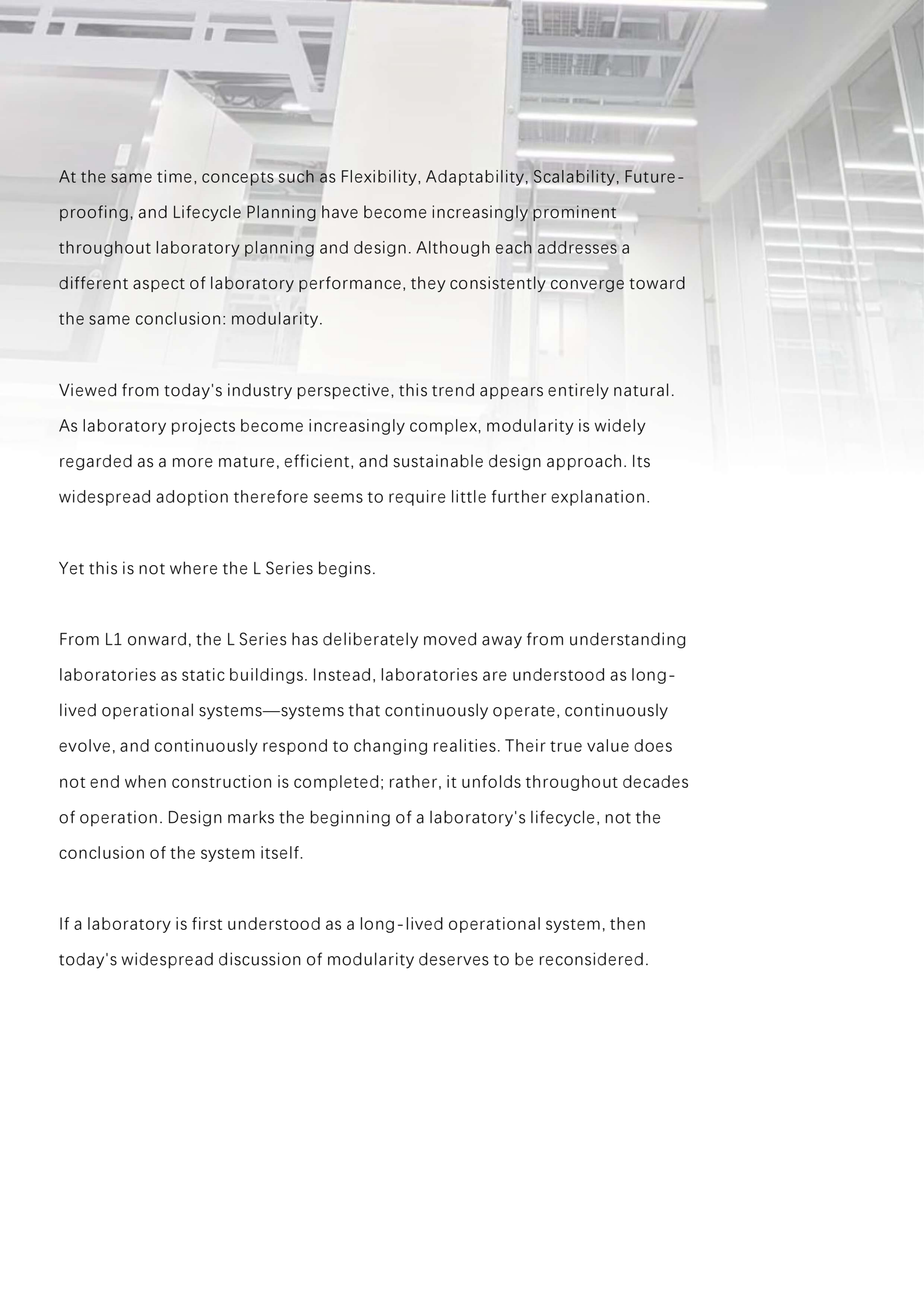
Is Laboratory Modularity the Goal of Design and Governance, or Its Result?

Yaowei Huang

1. A Consensus Rarely Questioned

In contemporary laboratory design, few concepts enjoy the level of acceptance that modularity does. Whether discussing new laboratory developments, facility renovations, long-term expansion, or lifecycle planning, modularity is almost universally regarded as the preferred direction for future development. It is valued not only for improving spatial efficiency, but also for its ability to accommodate future change, reduce renovation costs, maintain operational continuity, and extend facility lifespan. For an increasing number of laboratory projects, modularity is no longer simply one design strategy among many—it has gradually become the industry's default paradigm.

This trend is neither confined to a particular laboratory type nor driven by a specific design philosophy. From academic research laboratories and pharmaceutical R&D facilities to industrial laboratories and clinical testing environments, projects developed under entirely different regulatory frameworks, budget constraints, and technical requirements continue to exhibit remarkably similar organizational patterns. Repetitive functional units, standardized service infrastructure, regularized spatial organization, and expandable support systems appear repeatedly across otherwise unrelated projects.



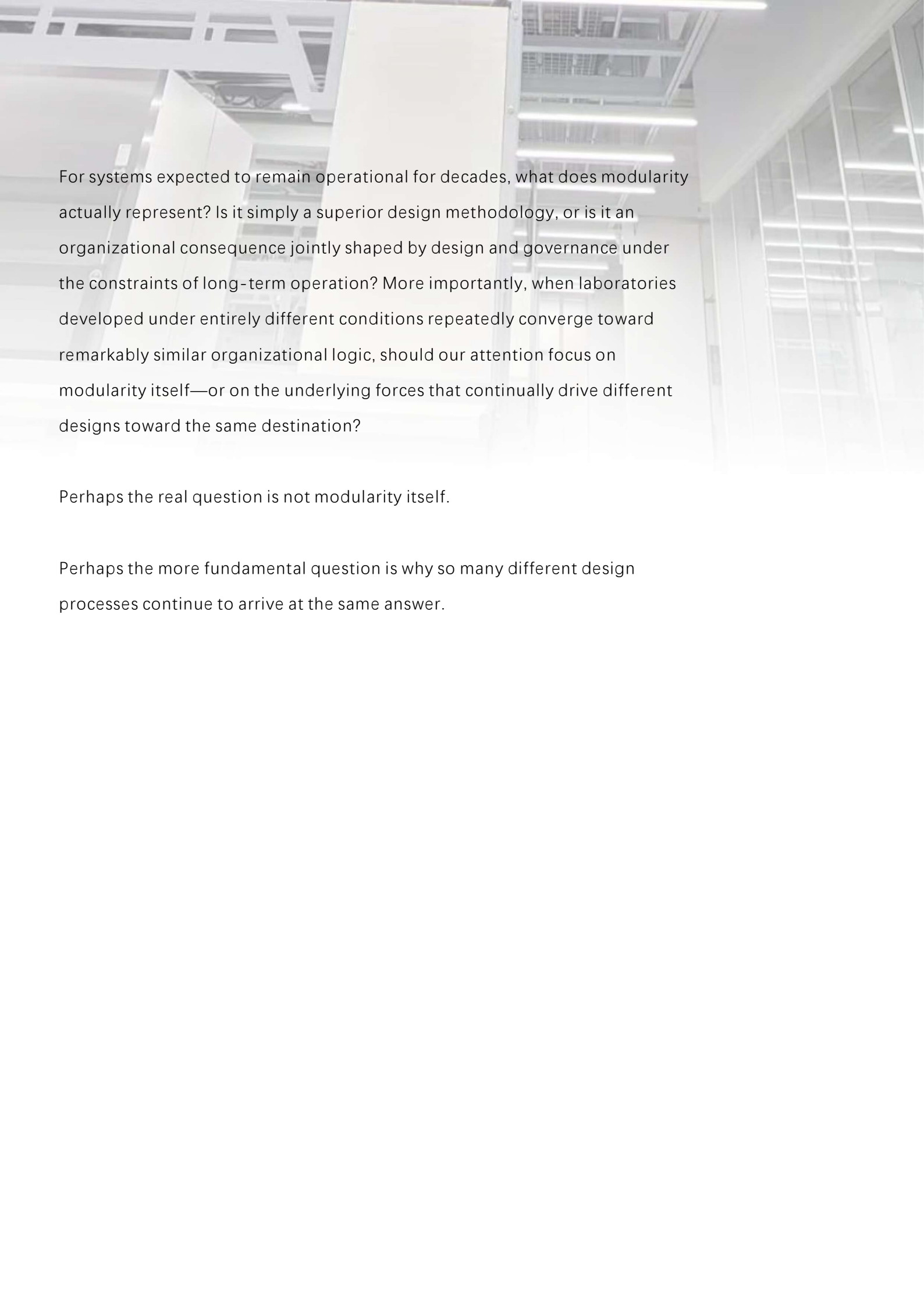
At the same time, concepts such as Flexibility, Adaptability, Scalability, Future-proofing, and Lifecycle Planning have become increasingly prominent throughout laboratory planning and design. Although each addresses a different aspect of laboratory performance, they consistently converge toward the same conclusion: modularity.

Viewed from today's industry perspective, this trend appears entirely natural. As laboratory projects become increasingly complex, modularity is widely regarded as a more mature, efficient, and sustainable design approach. Its widespread adoption therefore seems to require little further explanation.

Yet this is not where the L Series begins.

From L1 onward, the L Series has deliberately moved away from understanding laboratories as static buildings. Instead, laboratories are understood as long-lived operational systems—systems that continuously operate, continuously evolve, and continuously respond to changing realities. Their true value does not end when construction is completed; rather, it unfolds throughout decades of operation. Design marks the beginning of a laboratory's lifecycle, not the conclusion of the system itself.

If a laboratory is first understood as a long-lived operational system, then today's widespread discussion of modularity deserves to be reconsidered.



For systems expected to remain operational for decades, what does modularity actually represent? Is it simply a superior design methodology, or is it an organizational consequence jointly shaped by design and governance under the constraints of long-term operation? More importantly, when laboratories developed under entirely different conditions repeatedly converge toward remarkably similar organizational logic, should our attention focus on modularity itself—or on the underlying forces that continually drive different designs toward the same destination?

Perhaps the real question is not modularity itself.

Perhaps the more fundamental question is why so many different design processes continue to arrive at the same answer.

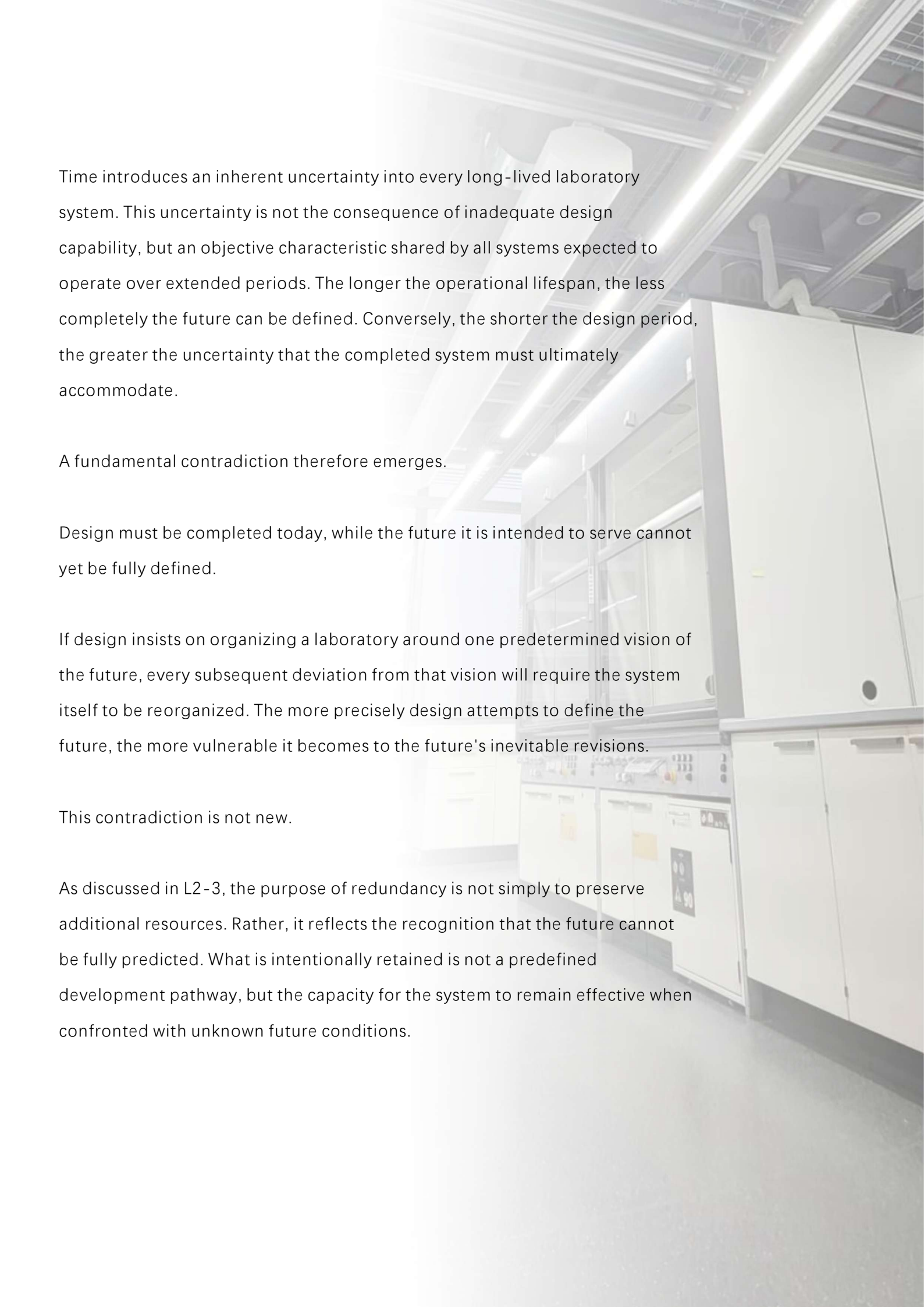
2. Why Can the Future Not Be Designed?

If modularity is not the original goal of laboratory design and governance, then a more fundamental question must first be answered: why do laboratory designs repeatedly converge toward modularity?

Answering this question requires looking beyond design methodology itself, because what ultimately reshapes design is not design—but time.

Every laboratory project begins at a specific moment in time. Within a limited design and construction period, decisions must be made regarding spatial organization, structural systems, equipment layouts, building services, and supporting infrastructure. Yet the environment these decisions are intended to serve does not exist at the moment they are made. It unfolds gradually over decades of operation. As time progresses, research priorities evolve, technologies advance, organizational structures change, equipment is replaced, and operational requirements continue to develop. Most of these changes cannot be fully anticipated during the design phase.

The fundamental question, therefore, is not whether change will occur, but how that change will unfold.



Time introduces an inherent uncertainty into every long-lived laboratory system. This uncertainty is not the consequence of inadequate design capability, but an objective characteristic shared by all systems expected to operate over extended periods. The longer the operational lifespan, the less completely the future can be defined. Conversely, the shorter the design period, the greater the uncertainty that the completed system must ultimately accommodate.

A fundamental contradiction therefore emerges.

Design must be completed today, while the future it is intended to serve cannot yet be fully defined.

If design insists on organizing a laboratory around one predetermined vision of the future, every subsequent deviation from that vision will require the system itself to be reorganized. The more precisely design attempts to define the future, the more vulnerable it becomes to the future's inevitable revisions.

This contradiction is not new.

As discussed in L2-3, the purpose of redundancy is not simply to preserve additional resources. Rather, it reflects the recognition that the future cannot be fully predicted. What is intentionally retained is not a predefined development pathway, but the capacity for the system to remain effective when confronted with unknown future conditions.

Time, therefore, changes more than the future itself.

It changes what design must define.

For a laboratory expected to operate over decades, design gradually shifts its attention away from defining a future spatial configuration and toward a more fundamental question: what should remain stable, and what must continue to accommodate change throughout the laboratory's lifecycle?

At this point, laboratory design begins to move beyond the organization of physical space and toward the organization of the system itself. The future may remain impossible to design directly, yet design can begin to establish the conditions that allow the system to continue responding to that future.

It is precisely through this transition that modularity begins to emerge.



3. Why Do Organizational Principles Ultimately Take the Form of Modules?

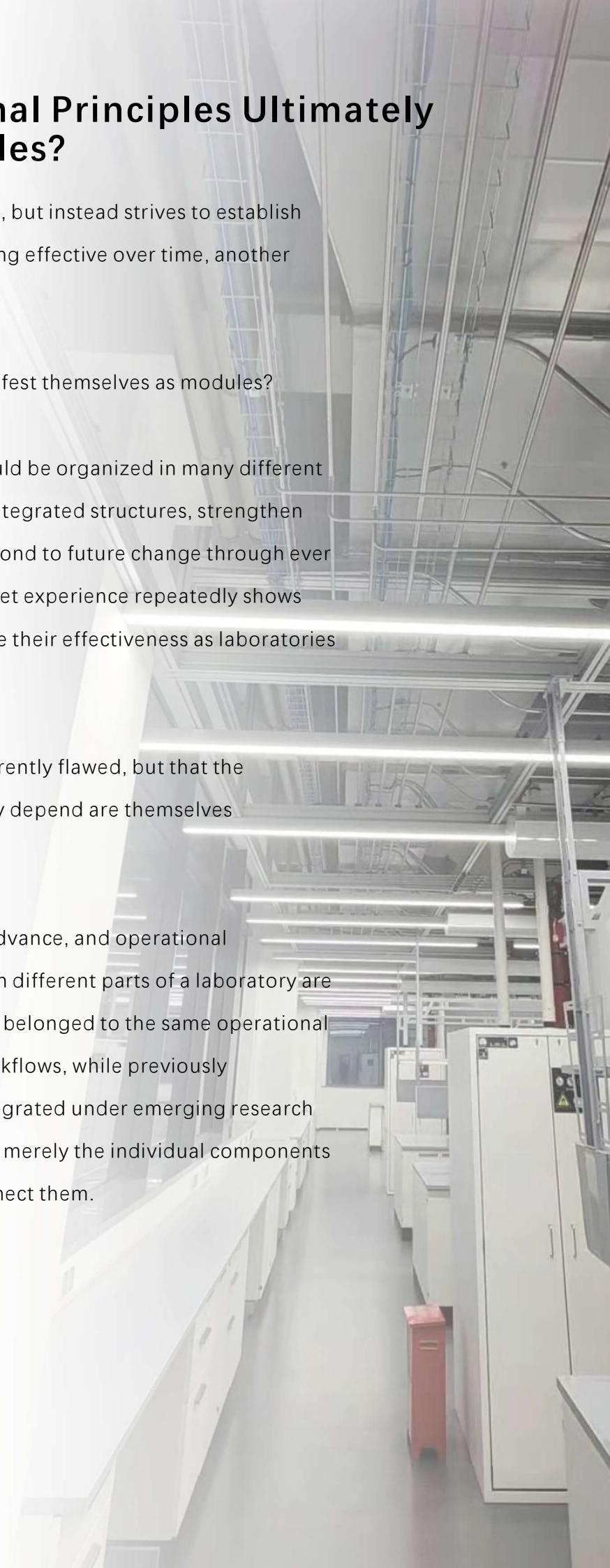
If design no longer seeks to define the future, but instead strives to establish organizational principles capable of remaining effective over time, another question naturally follows.

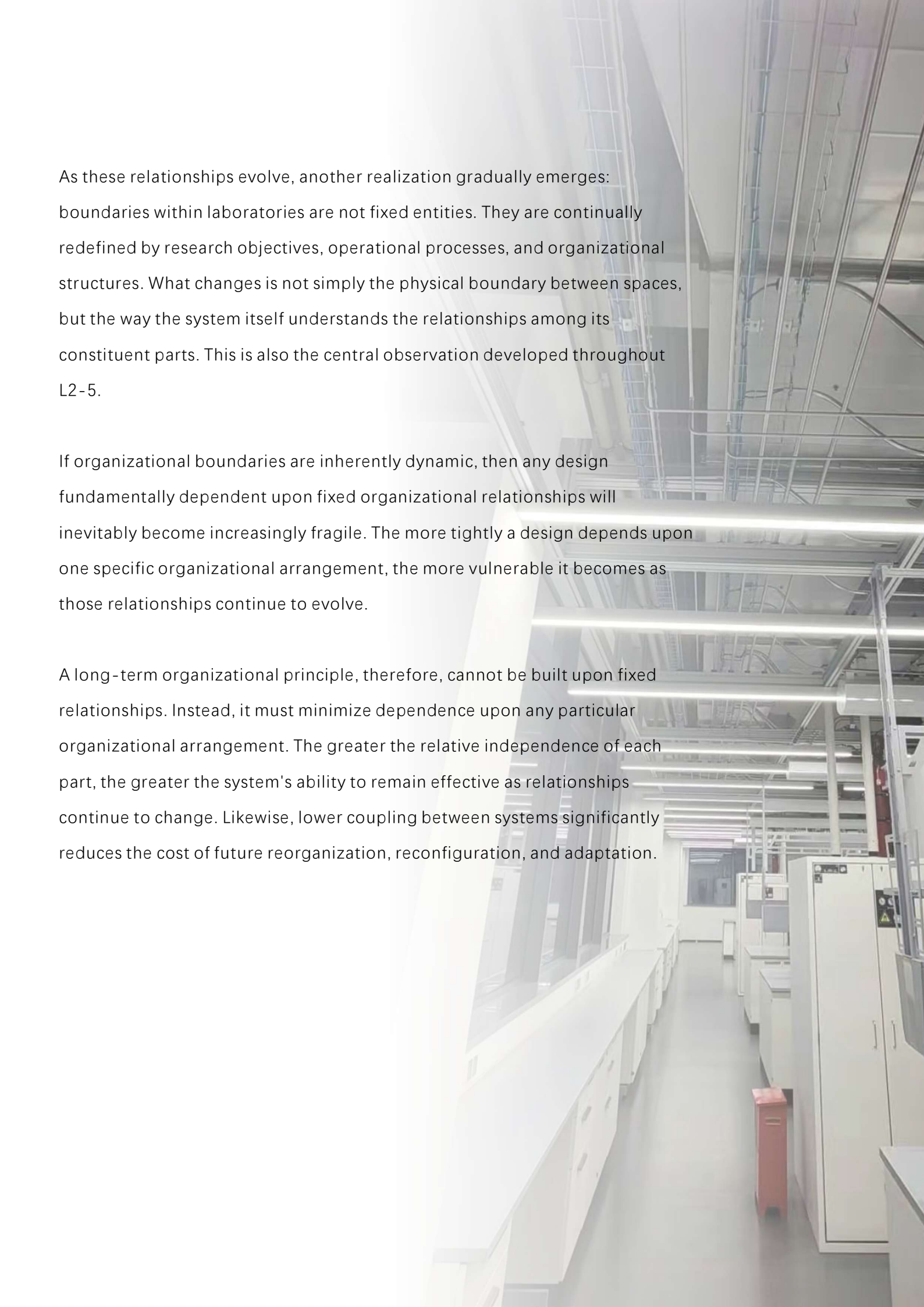
Why do these principles so consistently manifest themselves as modules?

In theory, a long-lived laboratory system could be organized in many different ways. It could rely on increasingly complex integrated structures, strengthen interdependencies between systems, or respond to future change through ever more elaborate organizational hierarchies. Yet experience repeatedly shows that many of these approaches gradually lose their effectiveness as laboratories continue to evolve.

The reason is not that these designs are inherently flawed, but that the organizational relationships upon which they depend are themselves continually changing.

As research priorities evolve, technologies advance, and operational requirements shift, the relationships between different parts of a laboratory are repeatedly reorganized. Functions that once belonged to the same operational unit may later become separated by new workflows, while previously independent spaces may become tightly integrated under emerging research programs. What changes continuously is not merely the individual components of the system, but the relationships that connect them.

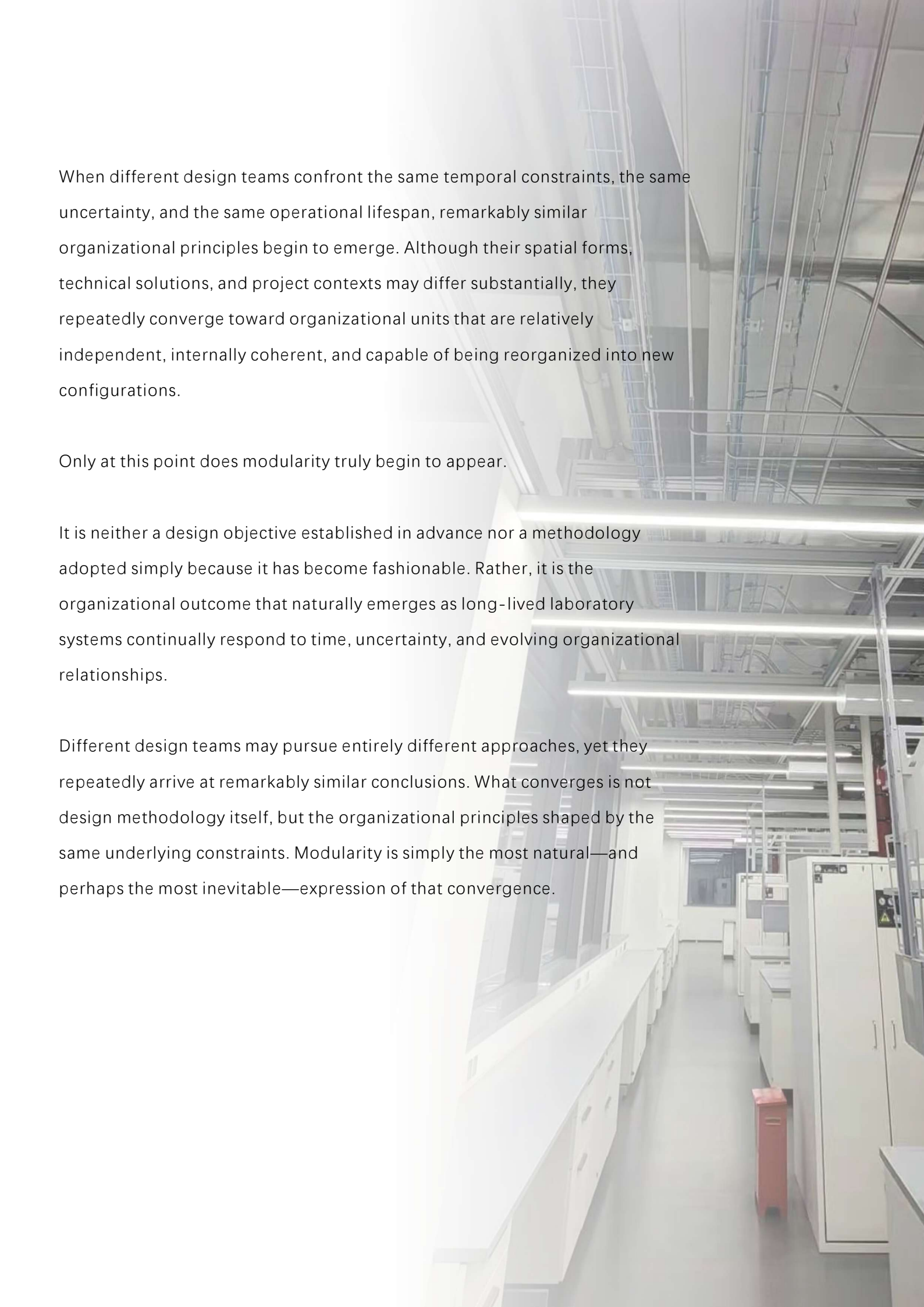




As these relationships evolve, another realization gradually emerges: boundaries within laboratories are not fixed entities. They are continually redefined by research objectives, operational processes, and organizational structures. What changes is not simply the physical boundary between spaces, but the way the system itself understands the relationships among its constituent parts. This is also the central observation developed throughout L2-5.

If organizational boundaries are inherently dynamic, then any design fundamentally dependent upon fixed organizational relationships will inevitably become increasingly fragile. The more tightly a design depends upon one specific organizational arrangement, the more vulnerable it becomes as those relationships continue to evolve.

A long-term organizational principle, therefore, cannot be built upon fixed relationships. Instead, it must minimize dependence upon any particular organizational arrangement. The greater the relative independence of each part, the greater the system's ability to remain effective as relationships continue to change. Likewise, lower coupling between systems significantly reduces the cost of future reorganization, reconfiguration, and adaptation.

A long, brightly lit laboratory hallway with white benches and equipment. The ceiling is high with visible metal trusses and numerous fluorescent light fixtures. On the left, there is a long white bench with a series of drawers underneath. On the right, there are tall white cabinets and a red fire extinguisher. The floor is a light gray. The overall atmosphere is clean and professional.

When different design teams confront the same temporal constraints, the same uncertainty, and the same operational lifespan, remarkably similar organizational principles begin to emerge. Although their spatial forms, technical solutions, and project contexts may differ substantially, they repeatedly converge toward organizational units that are relatively independent, internally coherent, and capable of being reorganized into new configurations.

Only at this point does modularity truly begin to appear.

It is neither a design objective established in advance nor a methodology adopted simply because it has become fashionable. Rather, it is the organizational outcome that naturally emerges as long-lived laboratory systems continually respond to time, uncertainty, and evolving organizational relationships.

Different design teams may pursue entirely different approaches, yet they repeatedly arrive at remarkably similar conclusions. What converges is not design methodology itself, but the organizational principles shaped by the same underlying constraints. Modularity is simply the most natural—and perhaps the most inevitable—expression of that convergence.

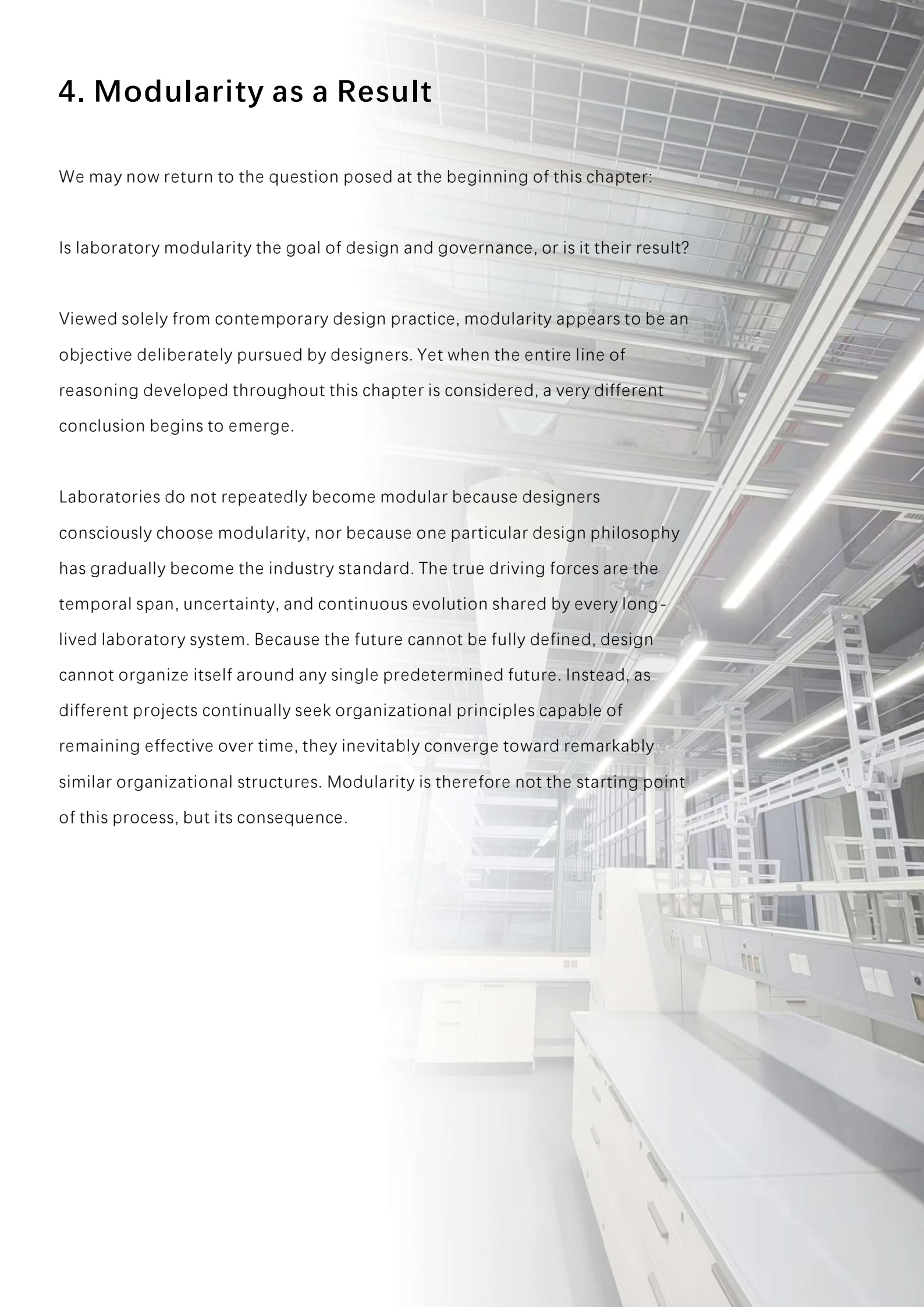
4. Modularity as a Result

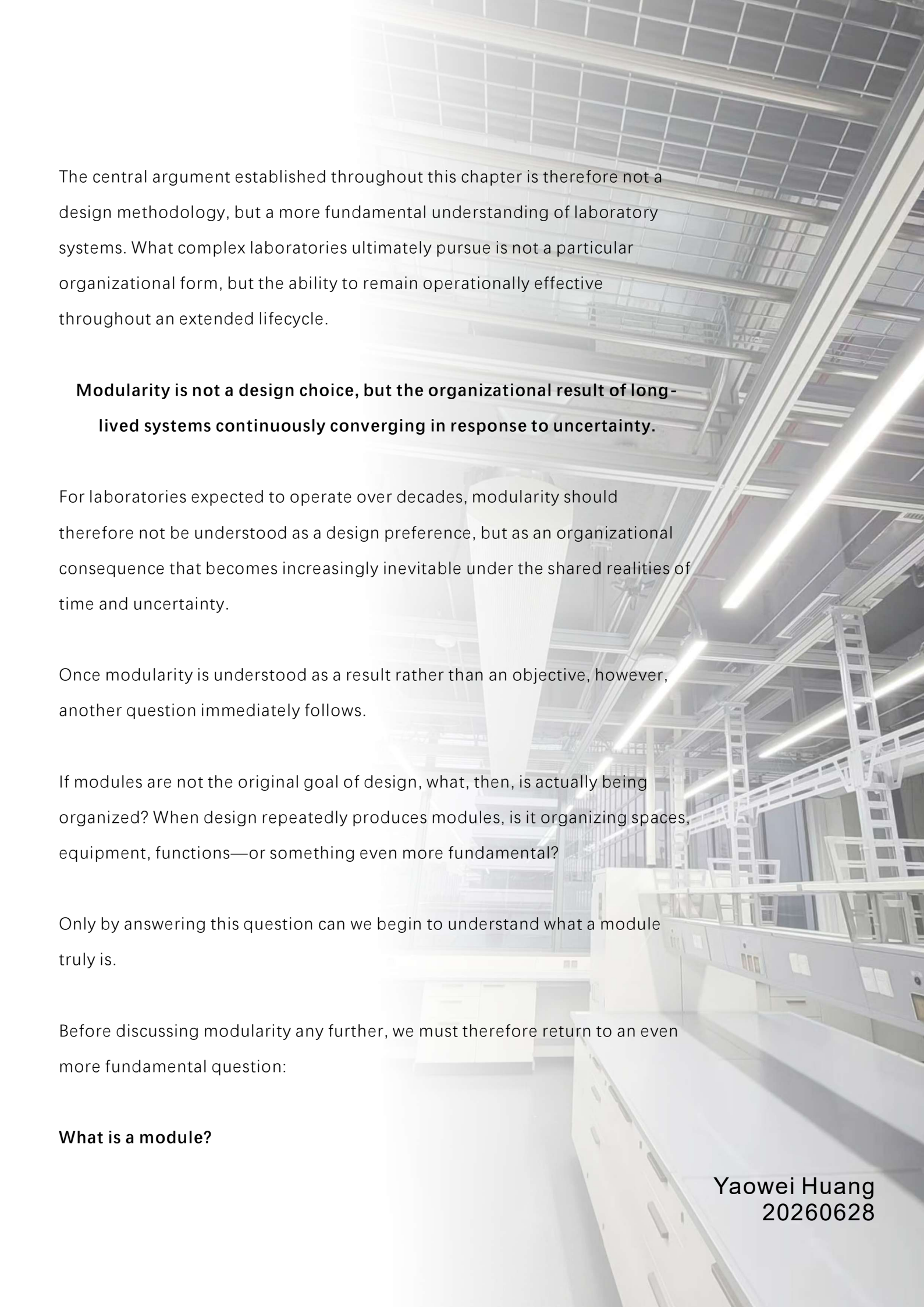
We may now return to the question posed at the beginning of this chapter:

Is laboratory modularity the goal of design and governance, or is it their result?

Viewed solely from contemporary design practice, modularity appears to be an objective deliberately pursued by designers. Yet when the entire line of reasoning developed throughout this chapter is considered, a very different conclusion begins to emerge.

Laboratories do not repeatedly become modular because designers consciously choose modularity, nor because one particular design philosophy has gradually become the industry standard. The true driving forces are the temporal span, uncertainty, and continuous evolution shared by every long-lived laboratory system. Because the future cannot be fully defined, design cannot organize itself around any single predetermined future. Instead, as different projects continually seek organizational principles capable of remaining effective over time, they inevitably converge toward remarkably similar organizational structures. Modularity is therefore not the starting point of this process, but its consequence.





The central argument established throughout this chapter is therefore not a design methodology, but a more fundamental understanding of laboratory systems. What complex laboratories ultimately pursue is not a particular organizational form, but the ability to remain operationally effective throughout an extended lifecycle.

Modularity is not a design choice, but the organizational result of long-lived systems continuously converging in response to uncertainty.

For laboratories expected to operate over decades, modularity should therefore not be understood as a design preference, but as an organizational consequence that becomes increasingly inevitable under the shared realities of time and uncertainty.

Once modularity is understood as a result rather than an objective, however, another question immediately follows.

If modules are not the original goal of design, what, then, is actually being organized? When design repeatedly produces modules, is it organizing spaces, equipment, functions—or something even more fundamental?

Only by answering this question can we begin to understand what a module truly is.

Before discussing modularity any further, we must therefore return to an even more fundamental question:

What is a module?

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